UAV DISASTER-ZONE STRATEGY ENGINE

Genetic Algorithm for Multi-Objective UAV Route Planning

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Abstract

This project implements a Genetic Algorithm (GA) to plan two unmanned aerial vehicles (UAVs) in a simulated 50 × 50 disaster grid. Each candidate solution encodes waypoint coordinates, flight speeds, and sensor power settings. A weighted fitness function rewards coverage of important targets while penalizing energy use, mission time, hazard exposure, and constraint violations. The implementation includes tournament selection, segment-based crossover, mutation, elitism, a population-diversity measure, random-candidate injection when diversity becomes low, and a random-search baseline. In the recorded run, the best GA fitness was 239.11, compared with a best random-search score of -73.21. Six sensitivity experiments were also performed.

1. Problem and Objectives

The task is a multi-objective optimization problem rather than a search for one predetermined route. The system must balance high-priority target detection against energy expenditure, flight time, and hazard exposure while respecting route-distance and no-fly-zone constraints. The simulated grid contains six targets with different priorities, five hazard locations, and restricted cells.

The project supports CLO4 by representing the operational world as structured knowledge—grid coordinates, target priorities, hazards, and restricted locations—and applying an inference/evaluation procedure to estimate the quality and safety of each proposed mission plan.

2. Chromosome Representation

A chromosome represents one complete fleet mission. It contains two UAV plans. Each UAV plan has eight waypoints, and every waypoint has an (x, y) coordinate, a speed value, and a sensor-power value. Coordinates are integers within the 50 × 50 grid; speed is bounded from 1.0 to 5.0; sensor power is bounded from 0.2 to 1.0. This mixed representation combines spatial decisions with continuous control settings in a form that can be changed by genetic operators.

Encoded component	Meaning	Allowed values
Waypoint (x, y)	Location visited by a UAV	Each coordinate 0–49
Speed	Movement speed for a waypoint segment	1.0–5.0
Sensor power	Sensor activation strength	0.2–1.0
Fleet chromosome	All plans for both UAVs	2 UAVs × 8 waypoints each

3. Fitness Function and Weighting

The fitness function combines weighted target detection with penalties for operational costs. For the implemented simulation, the expression is:

$$\text{Fitness} = 20 \times \text{weighted detection} - 0.03 \times \text{energy} - 0.10 \times \text{time} - 1.0 \times \text{hazard risk} - 100 \times \text{constraint violations}$$

Target detection is weighted by target importance: targets at (10,10), (20,35), and (40,40) have weight 3; targets at (35,15), (8,38), and (30,8) have weight 2. A sensor can contribute to detection when a waypoint lies within the sensor radius. Repeated detection contributions are combined so that coverage does not grow without bound. The coefficient 20 gives coverage a strong positive influence; energy, time, and hazard risk reduce fitness; and the large violation penalty discourages unsafe or infeasible routes. These coefficients are simulation design choices, not real-world safety constants.

4. Genetic Operators and Constraint Handling

The initial population is generated randomly. Tournament selection samples candidate chromosomes and chooses the one with the highest fitness. Segment-based crossover swaps matching sections of waypoint, speed, and sensor-power sequences between parent plans. Mutation may adjust a waypoint, speed, or sensor power while clamping values to their legal bounds. Elitism copies the two best chromosomes to the next generation. When normalized population diversity falls below 0.08, the algorithm injects fresh random chromosomes into approximately ten percent of the population to encourage exploration.

The route evaluator checks no-fly-zone intersections along sampled route segments, estimates distance, time, and energy, and adds risk when a waypoint is within five grid units of a hazard. Route-distance and no-fly-zone violations are handled through penalties in the fitness calculation. These checks make the optimizer prefer feasible routes, although a simulation result should not be treated as proof of real-world flight safety.

5. Experimental Results

The main recorded run used a population size of 40, 50 generations, mutation rate 0.15, and tournament size 3. The program saved convergence, route, coverage, baseline-comparison, and population-diversity visualizations in the outputs directory.

Main-run measure	Recorded result
Best Genetic Algorithm fitness	239.11
Best random-search fitness	-73.21
Average random-search fitness	-360.80
UAV 1 route distance	178.32
UAV 2 route distance	194.28
UAV 1 constraint violations	0
UAV 2 constraint violations	0

In this run, the best GA fitness was 312.32 points above the best random-search score. This difference suggests that evolutionary selection and variation found a substantially better candidate than the random-search sample set used in this experiment. The comparison is specific to this run and fitness definition; it does not prove that GA will outperform random search in every run. Fitness values can also be negative when penalties exceed the weighted detection reward.

6. Hyperparameter Sensitivity

Six experiments were run for 25 generations each. Population size, mutation rate, and tournament size (selection pressure) were varied around a reference setting. Results are shown below.

Exp.	Population	Mutation	Tournament	Best fitness
1	20	0.10	3	128.03
2	40	0.10	3	199.14
3	60	0.10	3	193.16
4	40	0.25	3	156.83
5	40	0.10	2	132.00
6	40	0.10	5	159.07

Among the tested settings, population 40, mutation 0.10, and tournament size 3 achieved the highest sensitivity-test fitness (199.14). Increasing population size from 20 to 40 improved the recorded score, but increasing it to 60 did not improve it further in this single trial. Raising mutation from 0.10 to 0.25 lowered the score in the tested setting, suggesting that excessive random change may disrupt useful solutions. Tournament size 2 and 5 both scored below size 3 in their respective tests. Because each setting was tested once, these trends should be treated as indicative, not statistically conclusive.

7. Interpretation and Limitations

The sensitivity results illustrate the trade-off between exploration and exploitation. A small population may explore fewer candidate plans, while a larger population costs more evaluations and does not guarantee a better outcome. Mutation introduces new variation, but too much can damage good partial routes. Tournament size affects selection pressure: larger tournaments tend to favor stronger candidates more often, which can speed improvement but may reduce variety. The diversity metric and random-candidate injection are intended to reduce premature convergence.

This is a synthetic demonstration. The hazard model, target priorities, energy/time equations, and fitness weights are simplified assumptions. The baseline comparison uses random search and the GA best score from one recorded run, rather than repeated statistical trials with confidence intervals. The sensitivity tests also use one run per configuration. Therefore, the results demonstrate the working design but do not establish a universally optimal parameter configuration.

The current coverage visualization represents the final solution's estimated sensor coverage, while the fitness-convergence plot records best and average fitness over generations. Further work could save coverage maps at multiple generations, repeat every experiment across multiple random seeds, report means and standard deviations, and improve the physical realism of flight and hazard models.

8. Conclusion

The UAV Strategy Engine demonstrates how a Genetic Algorithm can encode and optimize a fleet mission using waypoint locations, speed adjustments, and sensor-power profiles. Its object-oriented implementation includes a weighted fitness function, tournament selection, segment crossover, mutation, elitism, constraint penalties, diversity tracking, and a random-search comparison. The recorded main run reached a best fitness of 239.11 with zero reported constraint violations for both UAVs, outperforming the random-search baseline in that run. Sensitivity testing showed that parameter choices materially affected fitness. The system therefore provides a practical example of knowledge representation and evaluation for an AI optimization problem, while remaining a simulation rather than a deployable flight-safety system.

Files and Reproducibility

Run the simulation from the project environment with `python main.py`. The script writes generated visualizations and `sensitivity_results.csv` to the `outputs/` directory. The report's numerical results reflect the terminal output provided for the recorded run; a later run may differ if random state or code settings change.

Link to GitHub Repository: *https://github.com/abdullah-dex/UAV-Strategy-Engine.git*